

Toxicogenomics I

Linking pollution to functional genomics

Dr. Marie Brasseur, AG Ecotoxicology

What are Toxicogenomics?

- *How environmental pollutants affect the genome of living organisms and its expression dynamics*
- Identify the molecular pathways underlying biological responses that are affected/induced by toxicant exposure
- Important to gain **mechanistic insights**

The central dogma of molecular biology



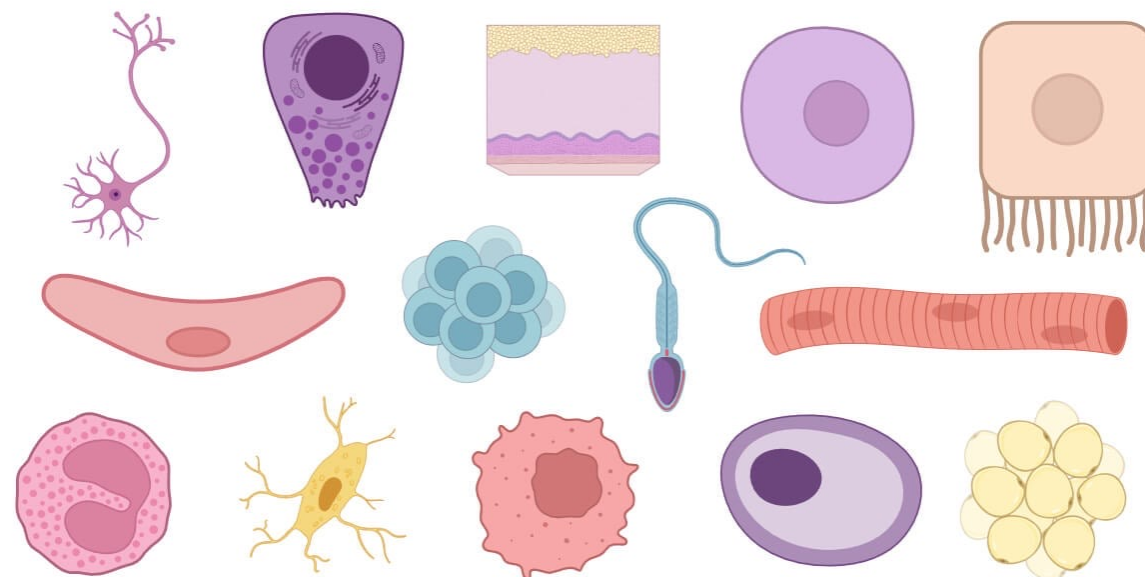
Different cell types have the same genotype...

Genotype

- Encodes the genetic information
- Static

Phenotype

- Expression of genetic information
- Plastic



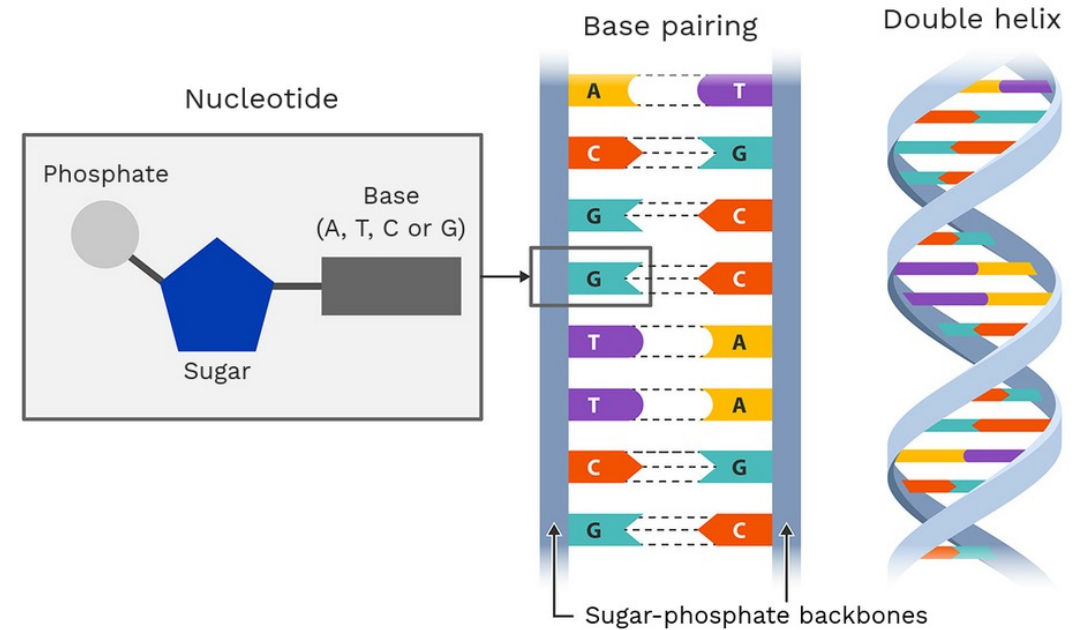
... but a very different phenotype

The central dogma of molecular biology



Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)

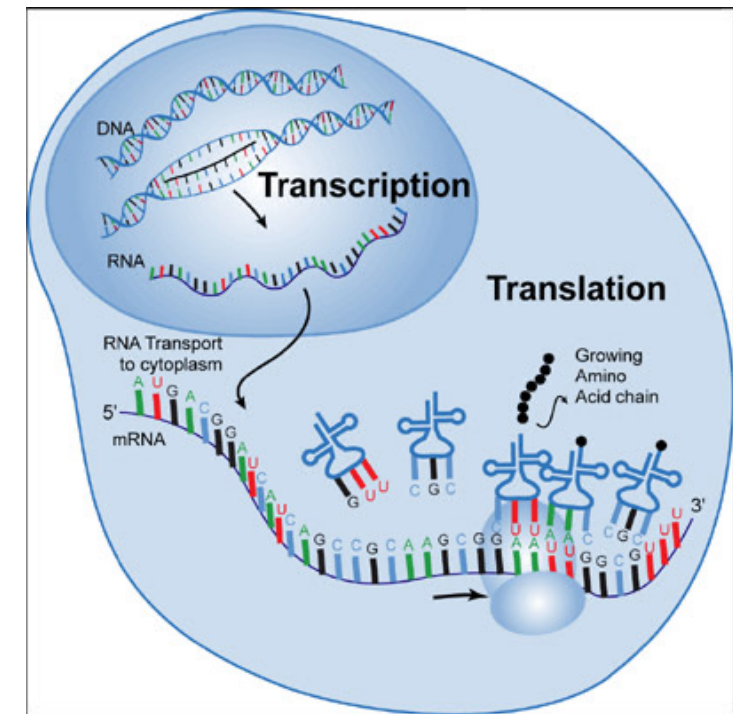


The central dogma of molecular biology



Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)
- Transcription: DNA (double stranded) -> RNA (single stranded) via complementary base pairing

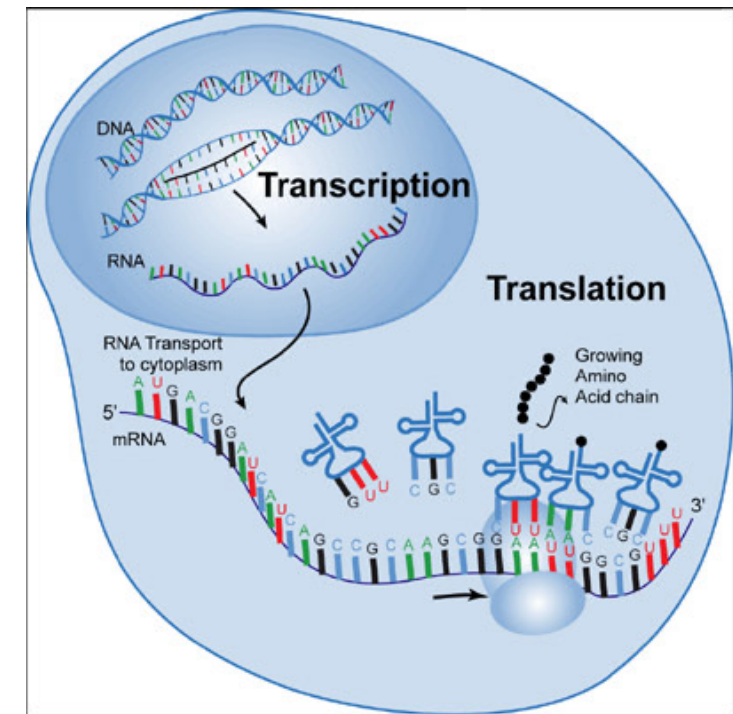


The central dogma of molecular biology



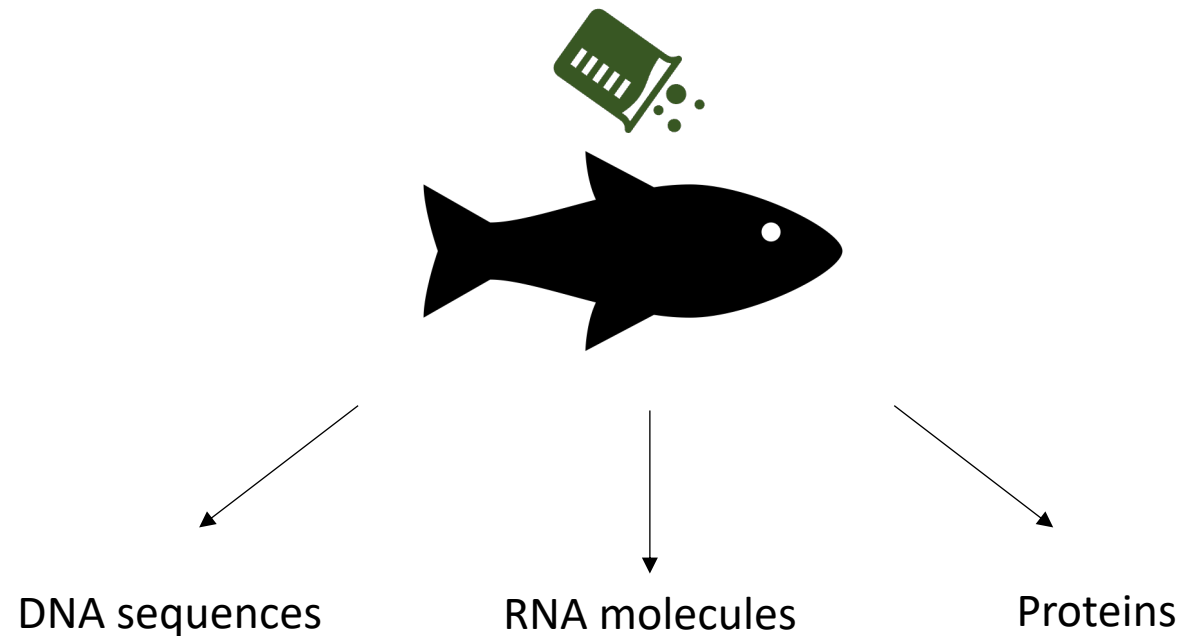
Encoding of genetic information:

- Sequence of bases
- Complementary base pairing (A-T & C-G)
- Transcription: DNA (double stranded) -> RNA (single stranded) via complementary base pairing
- Translation: RNA triplet -> amino acid



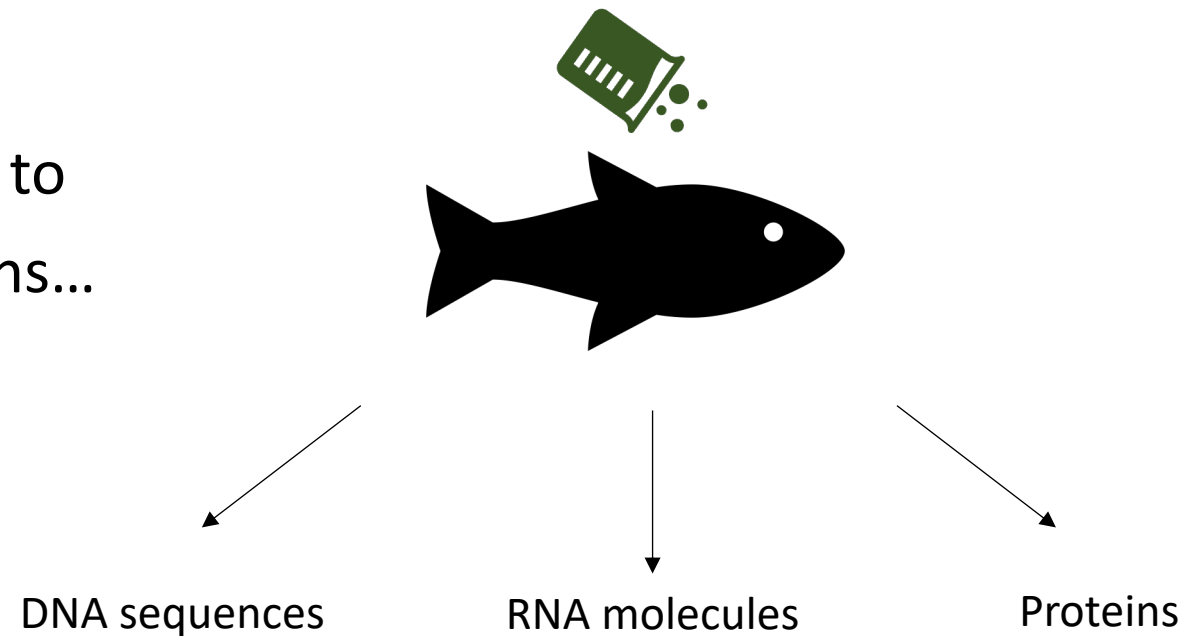
What are Toxicogenomics?

- *How environmental pollutants affect the genome of living organisms and its expression dynamics*



What are Toxicogenomics?

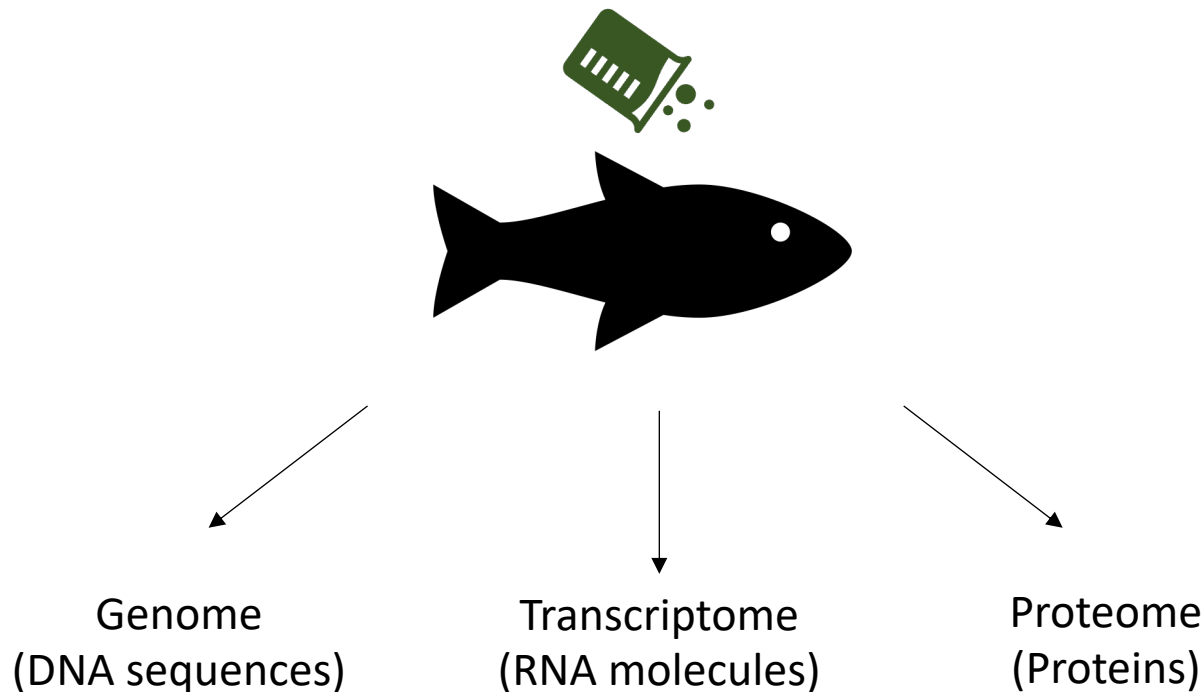
- *How environmental pollutants affect the genome of living organisms and its expression dynamics*
- While we were earlier limited to single genes, transcripts, proteins...



What are Toxicogenomics?

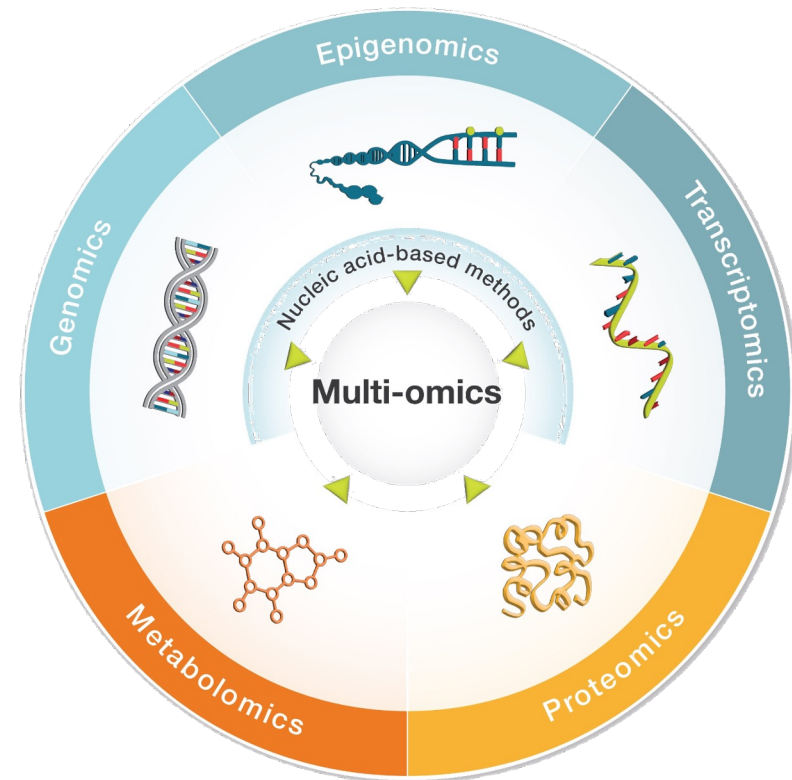
- *How environmental pollutants affect the genome of living organisms and its expression dynamics*

...we can nowadays assess whole genomes, transcriptomes, proteomes (metabolomes, epigenomes, lipidomes...)



What are Toxicogenomics?

- *How environmental pollutants affect the (epi)genome, the transcriptome, the proteome and the metabolome of living organisms*
- Integration of **–OMICS** in ecotoxicology



Recap questions

- What is stated by the central dogma of molecular biology?
- How is genetic information encoded?
- How can we infer the (unknown) RNA sequence from a (known) DNA sequence?
- What is the difference between the genome and the transcriptome?

The technical part: -OMICS

- High-throughput technologies to measure

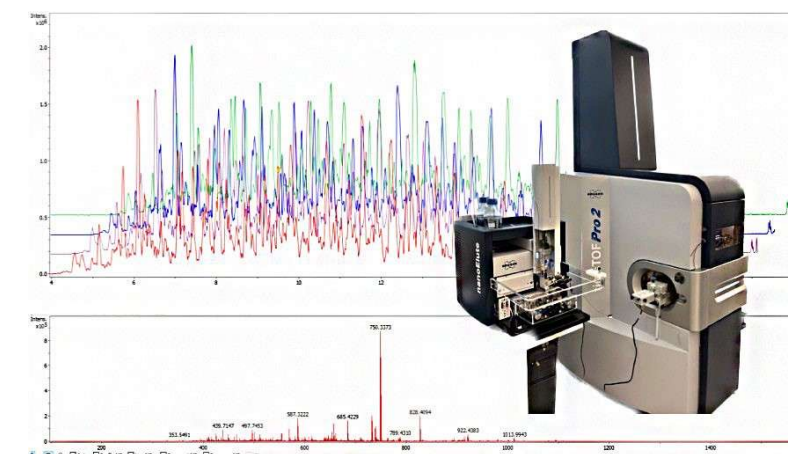
- (Epi-)Genome, Transcriptome

→ Based on next generation sequencing
(decode base sequence of DNA & RNA)

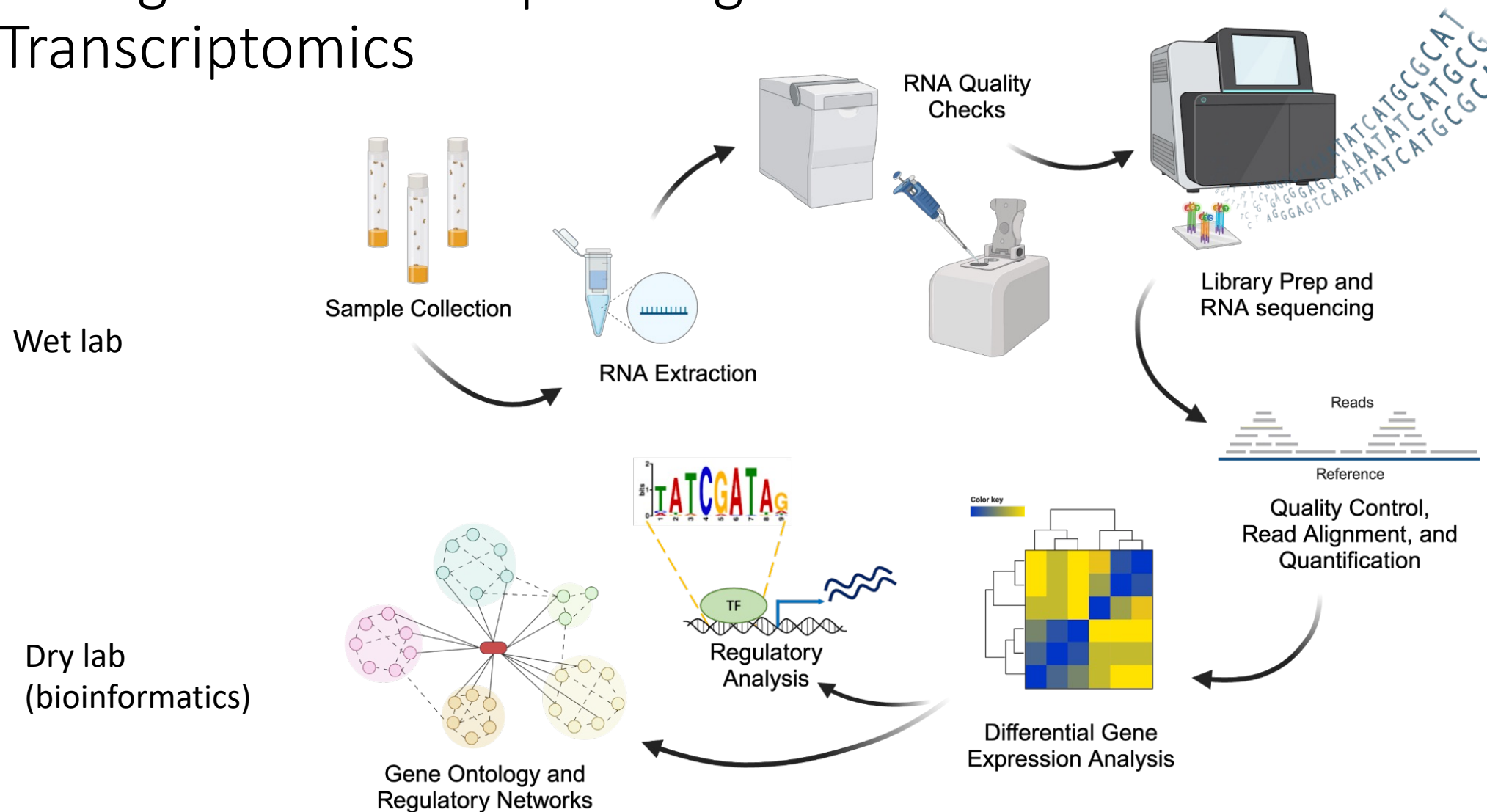


- Proteome, Metabolome, Lipidome

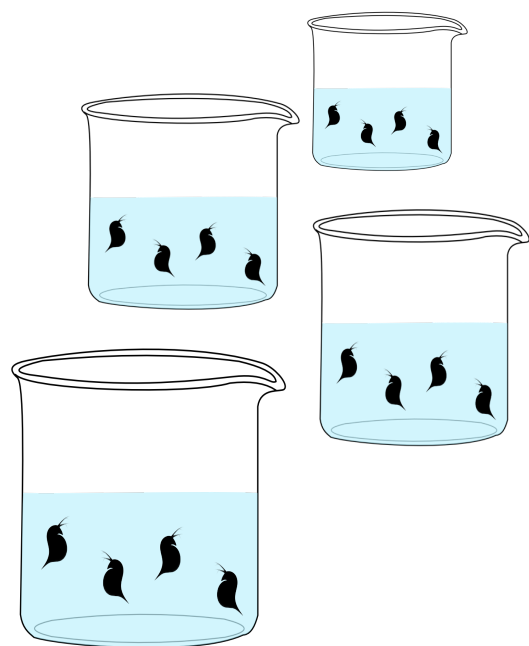
→ Based on mass spectrometry
(mass spectrum = compound specific
pattern of mass/charge ratio peaks;
not covered here)



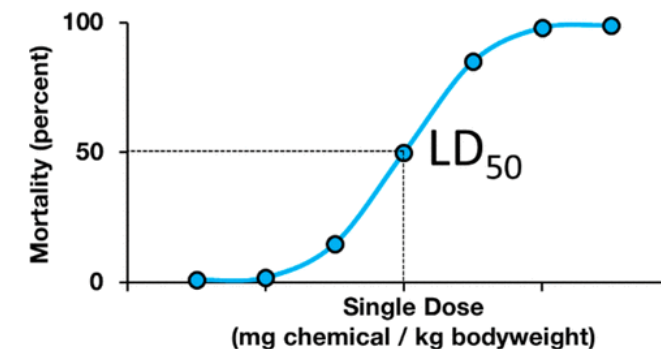
Next generation sequencing of RNA: Transcriptomics



Wet lab



Measure e.g. survival



Measure e.g. gene expression

- Pre-OMICS:
 - qPCR -> single genes
 - Microarrays -> multiple genes (hundreds to thousands)
- -OMICS
 - RNA sequencing -> global assessment of the whole transcriptome

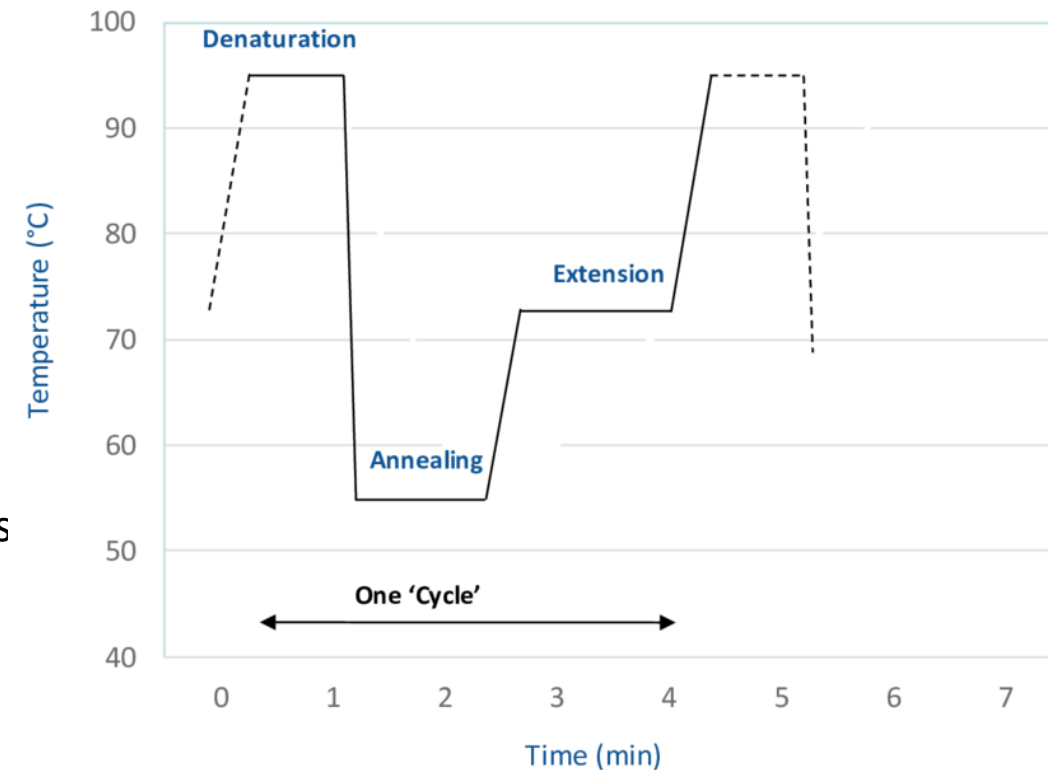
(Eco)toxicological experiment

Wet lab – (q)PCR

- PCR: Polymerase chain reaction
- Artificial DNA replication in 3 steps:
 - PCR ingredients: DNA polymerase (e.g. *Taq*), nucleotides (dNTPs), primers, template DNA

One cycle:

1. Heat denaturation: double-stranded DNA → 2 single strands
2. Primer annealing
3. Extension: DNA polymerase can bind to primer and extends the single stranded DNA (template) by synthesizing the complementary DNA strand



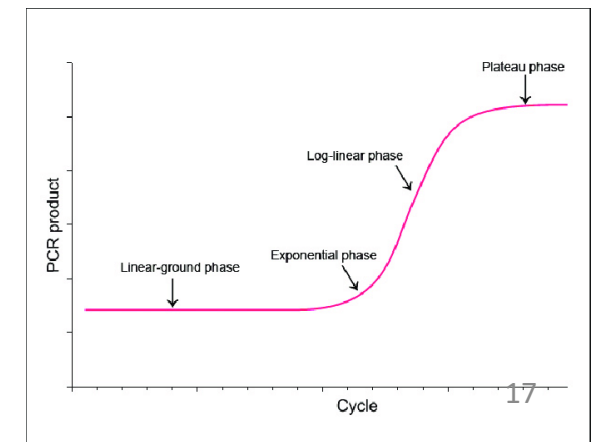
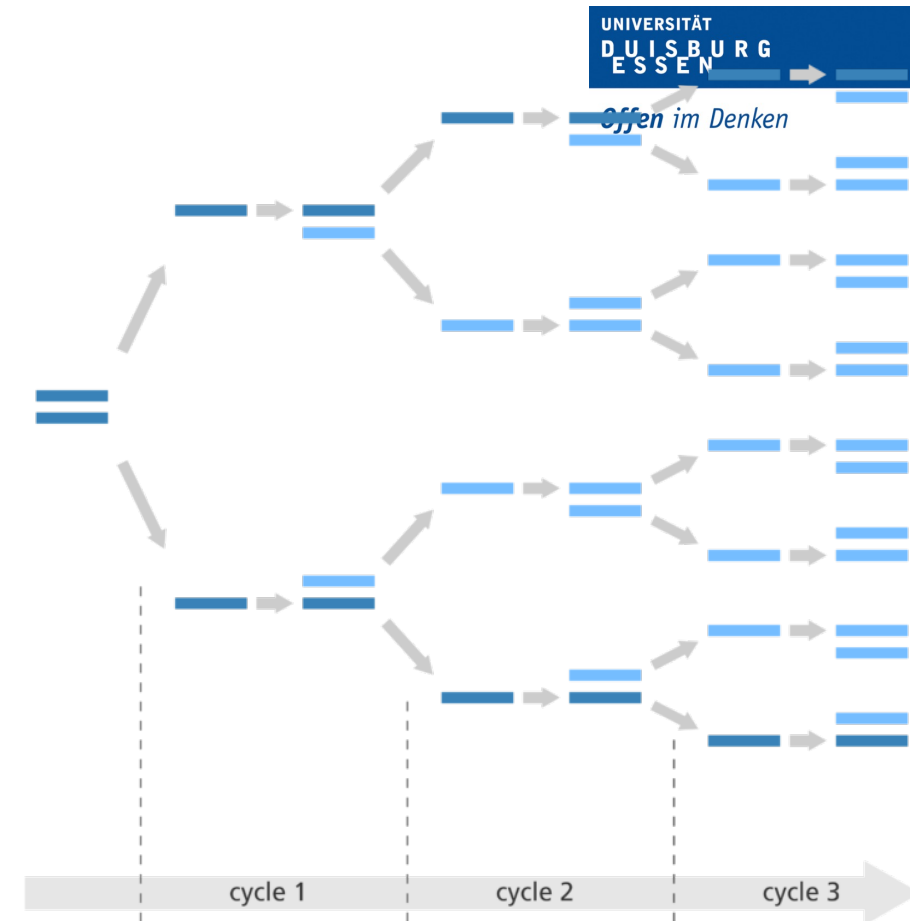
Wet lab – (q)PCR

- PCR: Polymerase chain reaction
- Artificial DNA replication in 3 steps:
 - PCR ingredients: DNA polymerase (e.g. *Taq*), nucleotides (dNTPs), primers, template DNA

One cycle:

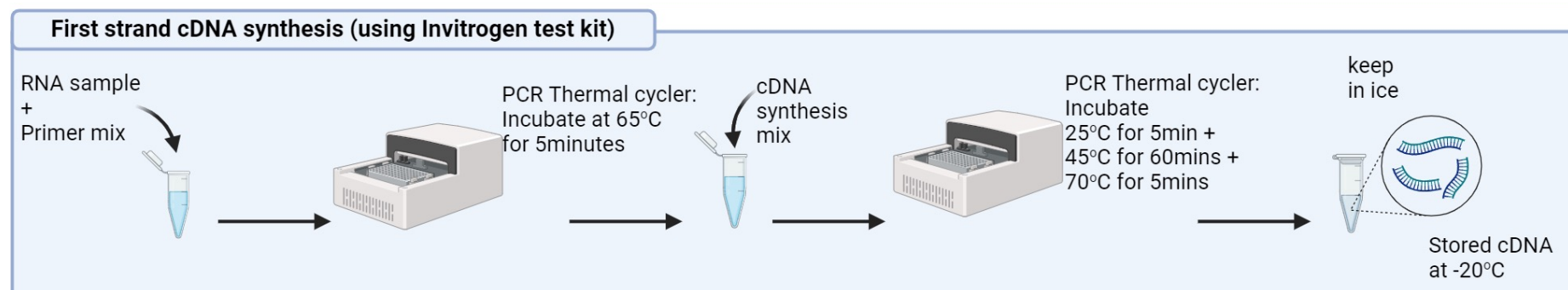
1. Heat denaturation: double-stranded DNA → 2 single strands.
2. Primer annealing
3. Extension: DNA polymerase can bind to primer and extends the single stranded DNA (template) by synthesizing the complementary DNA strand

$\sim 2^n$ copies (n =cycle number)

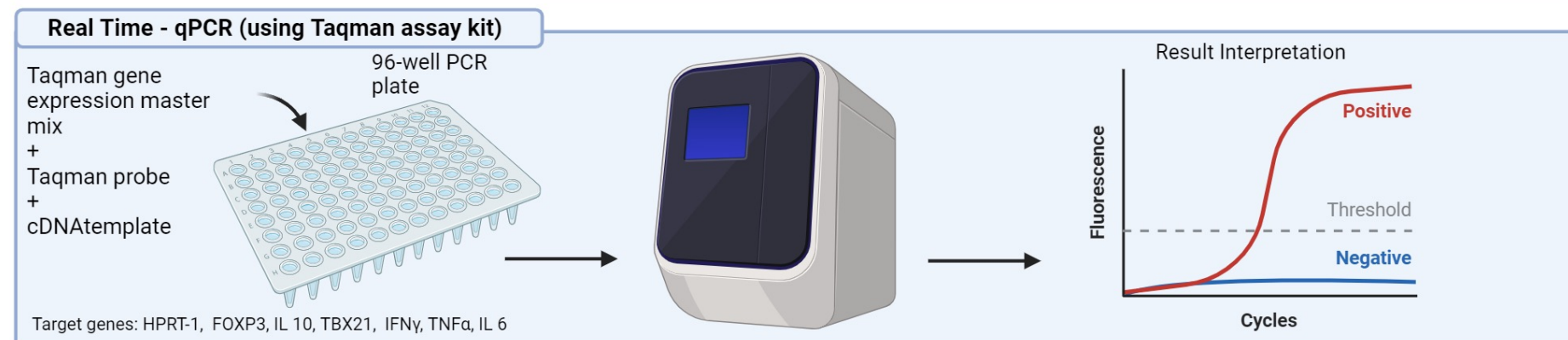


Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction



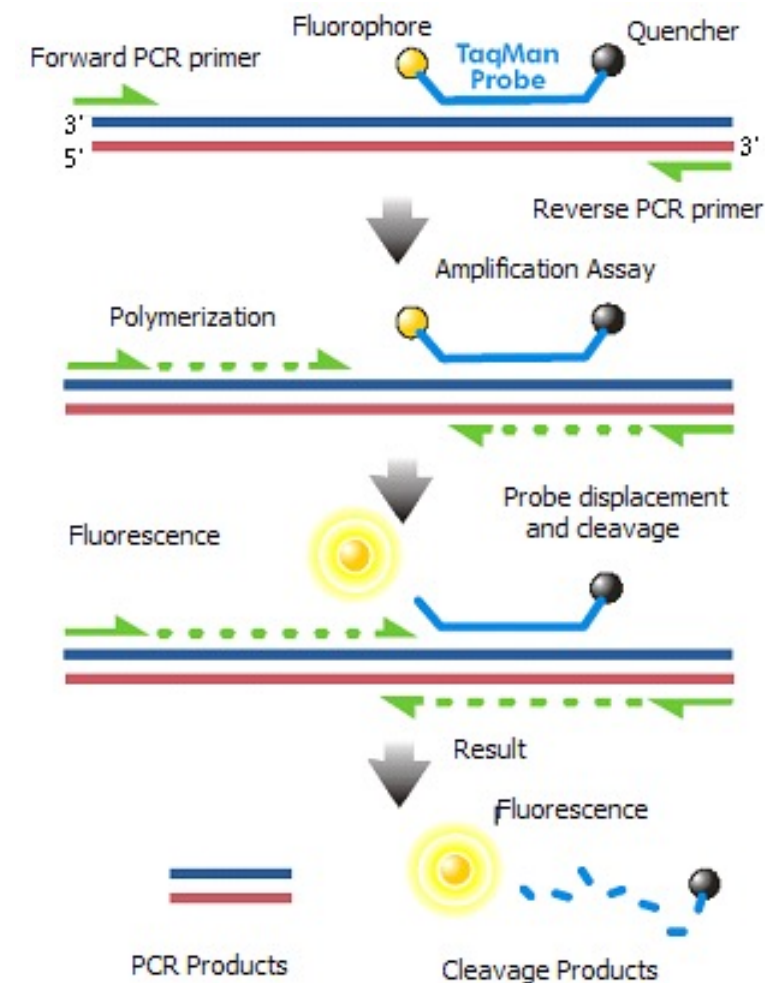
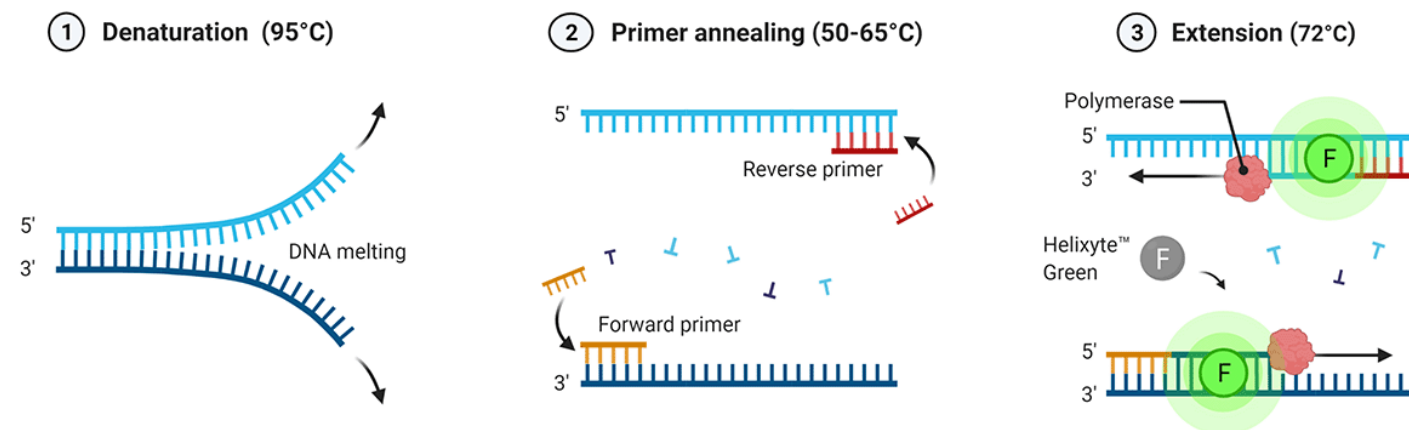
RNA must be
reversely transcribed



Polymerase reaction

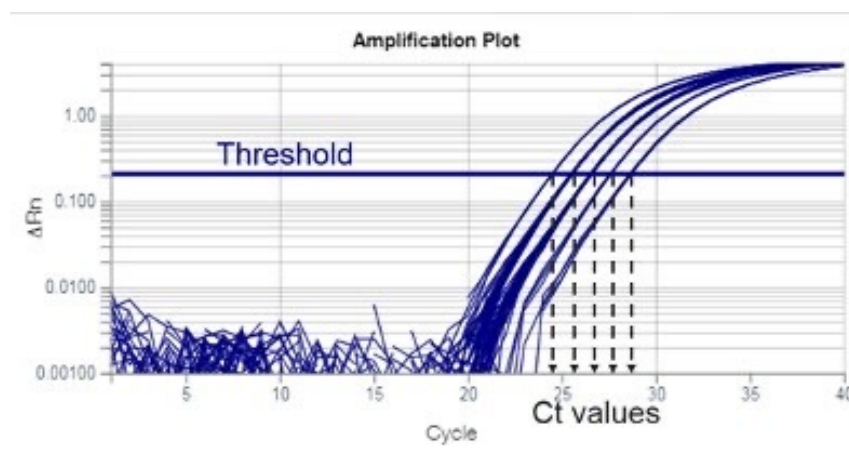
Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction
 - During PCR based amplification, light signal is emitted

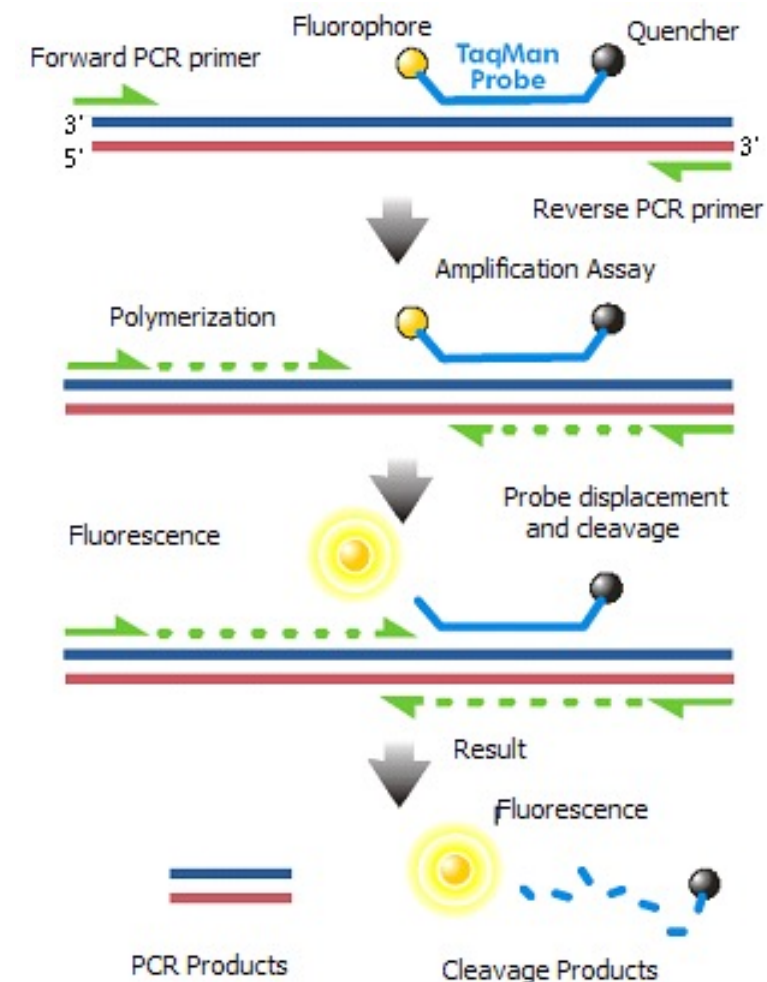


Wet lab – (q)PCR

- qPCR: quantitative Polymerase chain reaction
 - During PCR based amplification, light signal is emitted



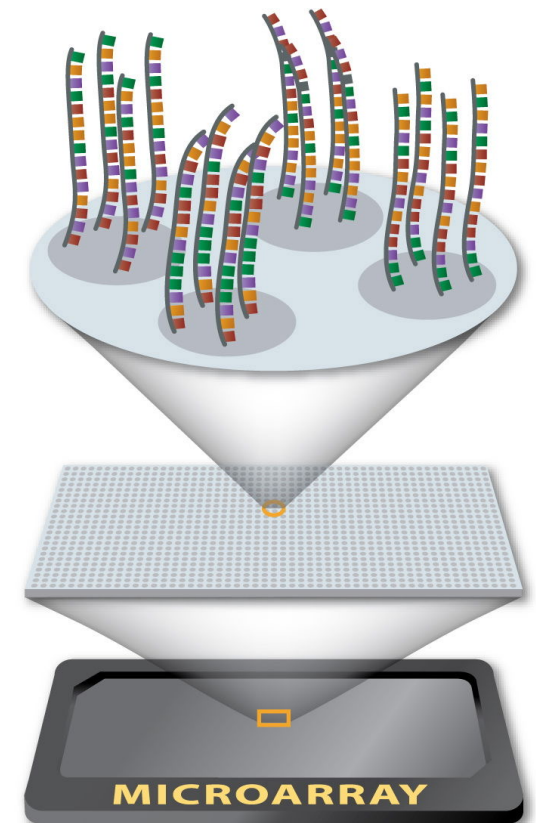
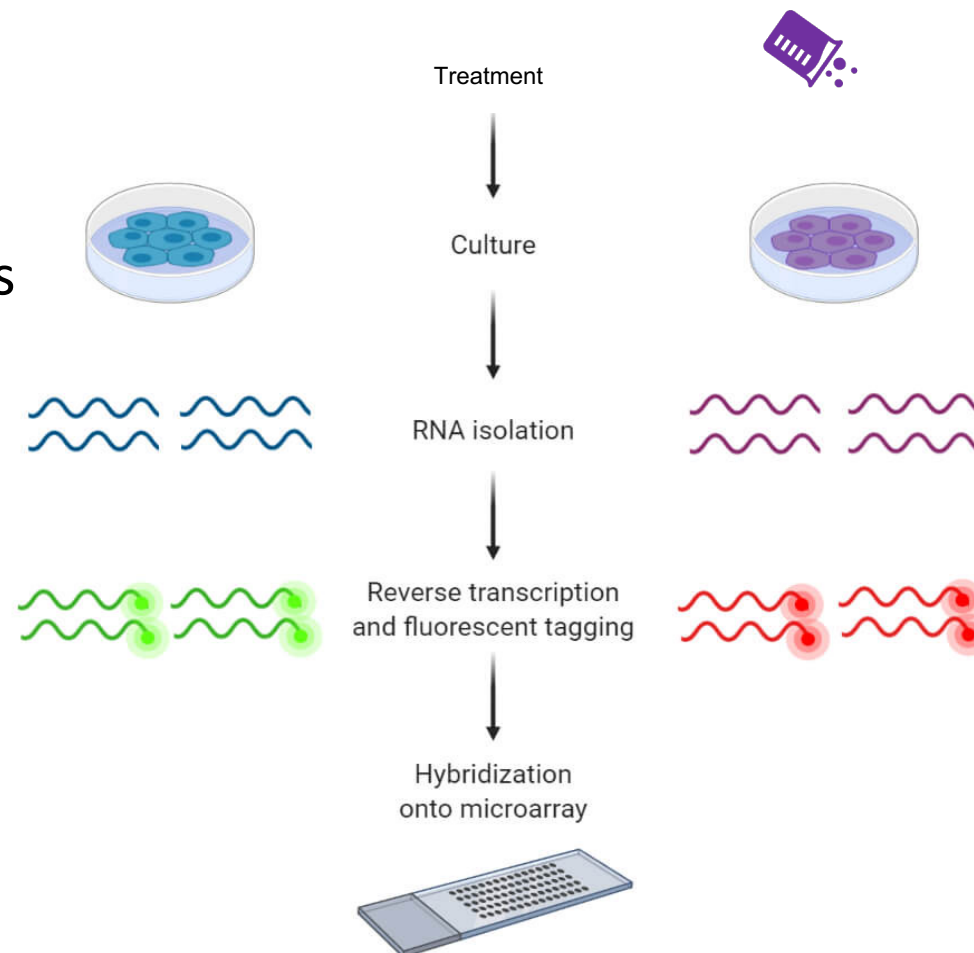
$$\text{Quantity} \sim e^{-Ct}$$



Wet lab – cDNA microarrays

Microarrays

- High throughput
(hundreds to thousands
of genes)



Wet lab – cDNA microarrays

Microarrays

- High throughput
(hundreds to thousands
of genes)
- Still prior knowledge about the
genome/transcriptome required
for probe design

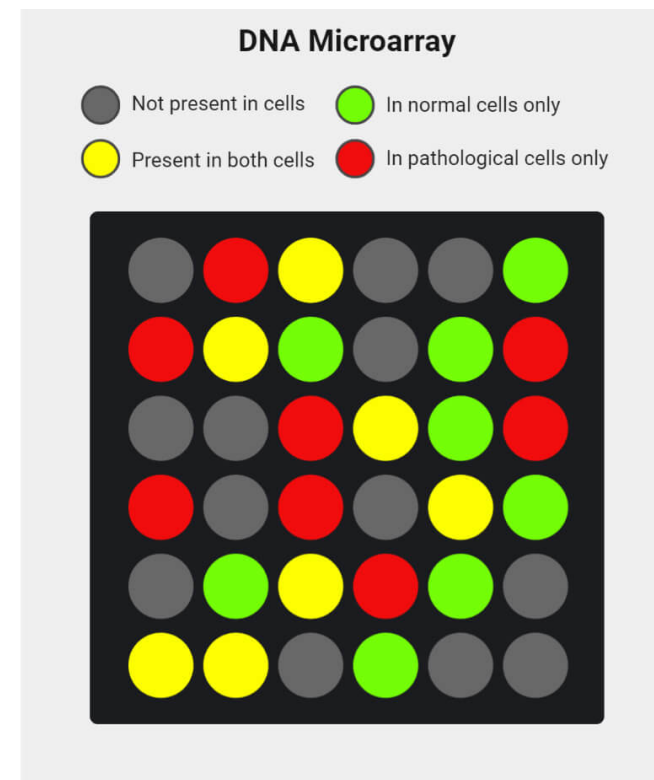
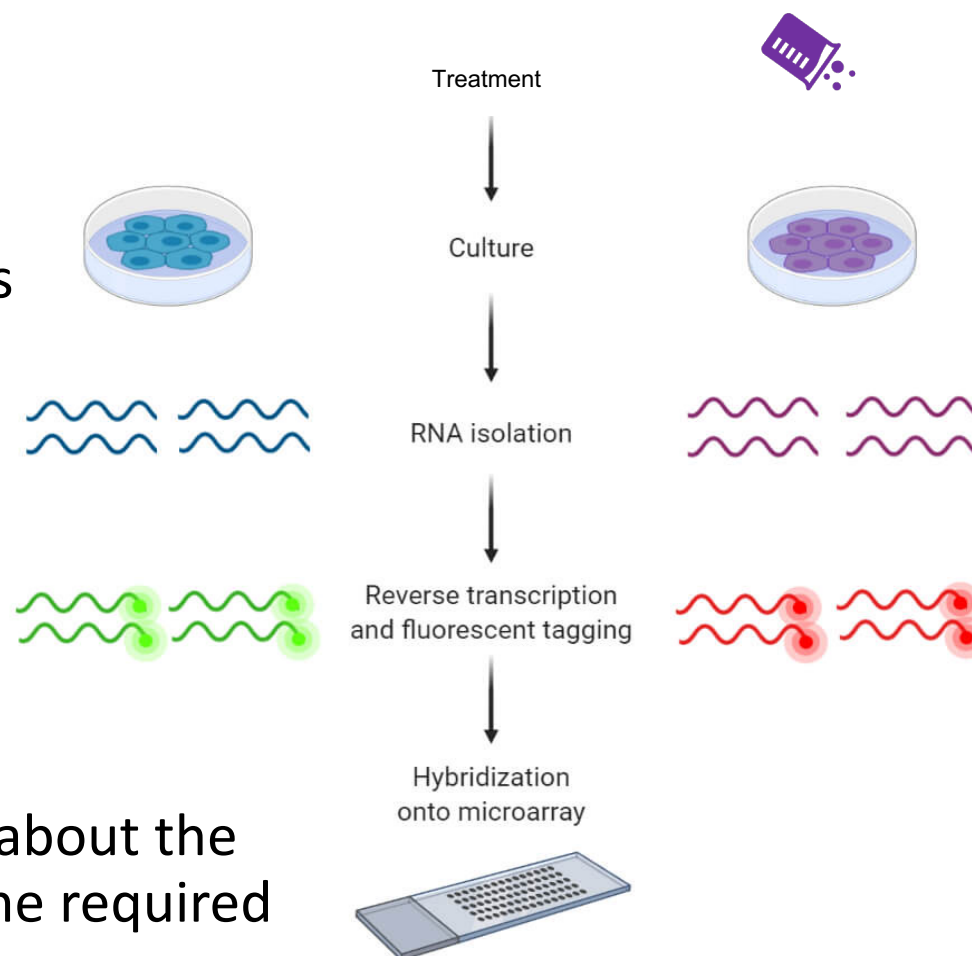
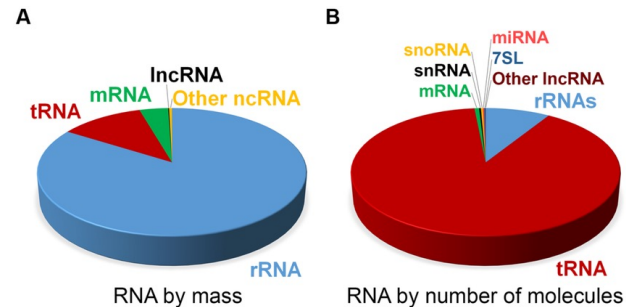


Image By Sagar Aryal, created using biorender.com

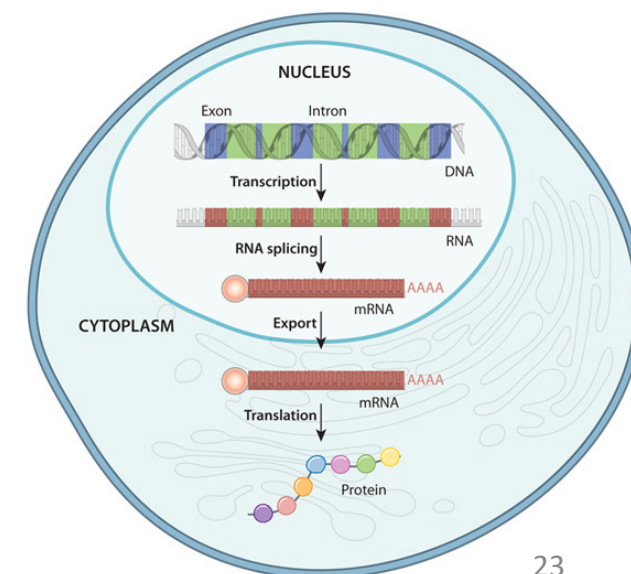
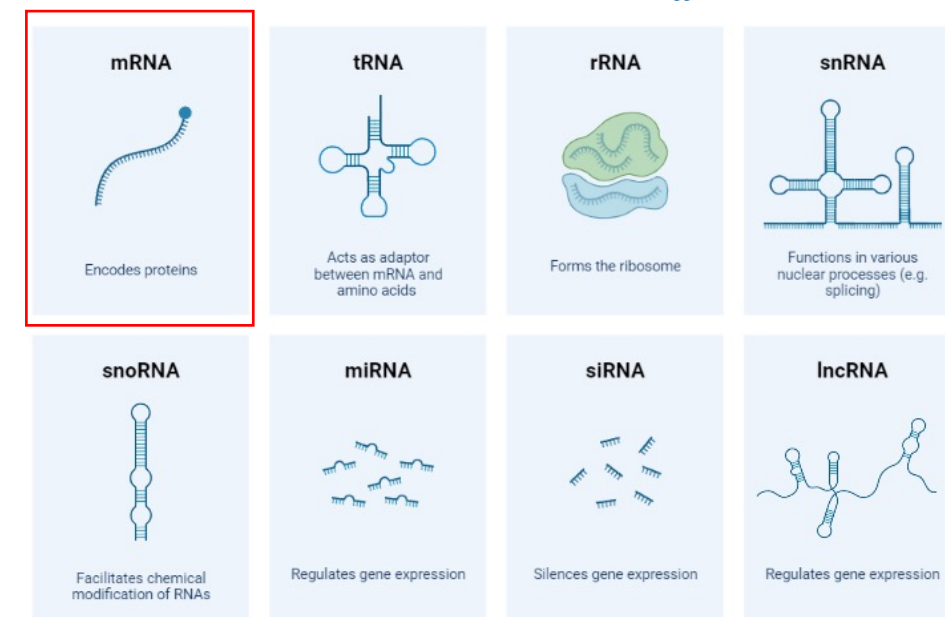
Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA



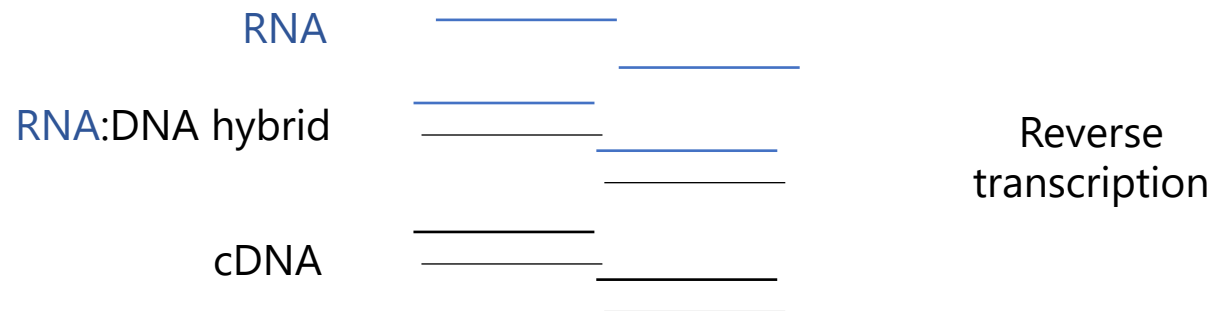
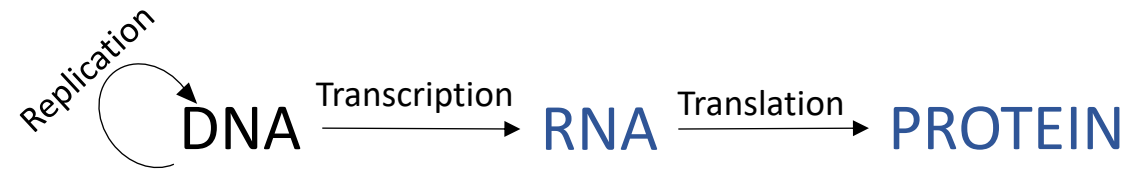
Poly(A) selection



Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

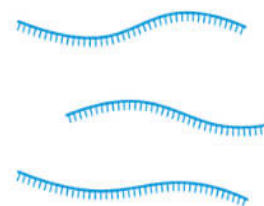


Wet lab - RNA-Seq

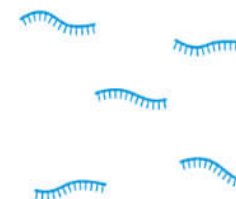
RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

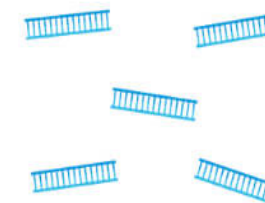
① Isolate RNA from samples



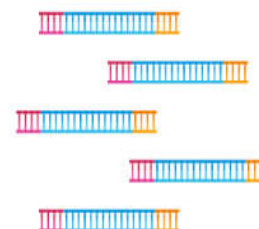
② Fragment RNA into short segments



③ Convert RNA fragments into cDNA



④ Ligate sequencing adapters and amplify



⑤ Perform NGS sequencing



Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

2 videos recommended:

<https://www.youtube.com/watch?v=CZeN-IgjYCo>

Or

<https://www.youtube.com/watch?v=EDVKxSNdSic>

(from Illumina; more comple)

Next Gen Sequencing

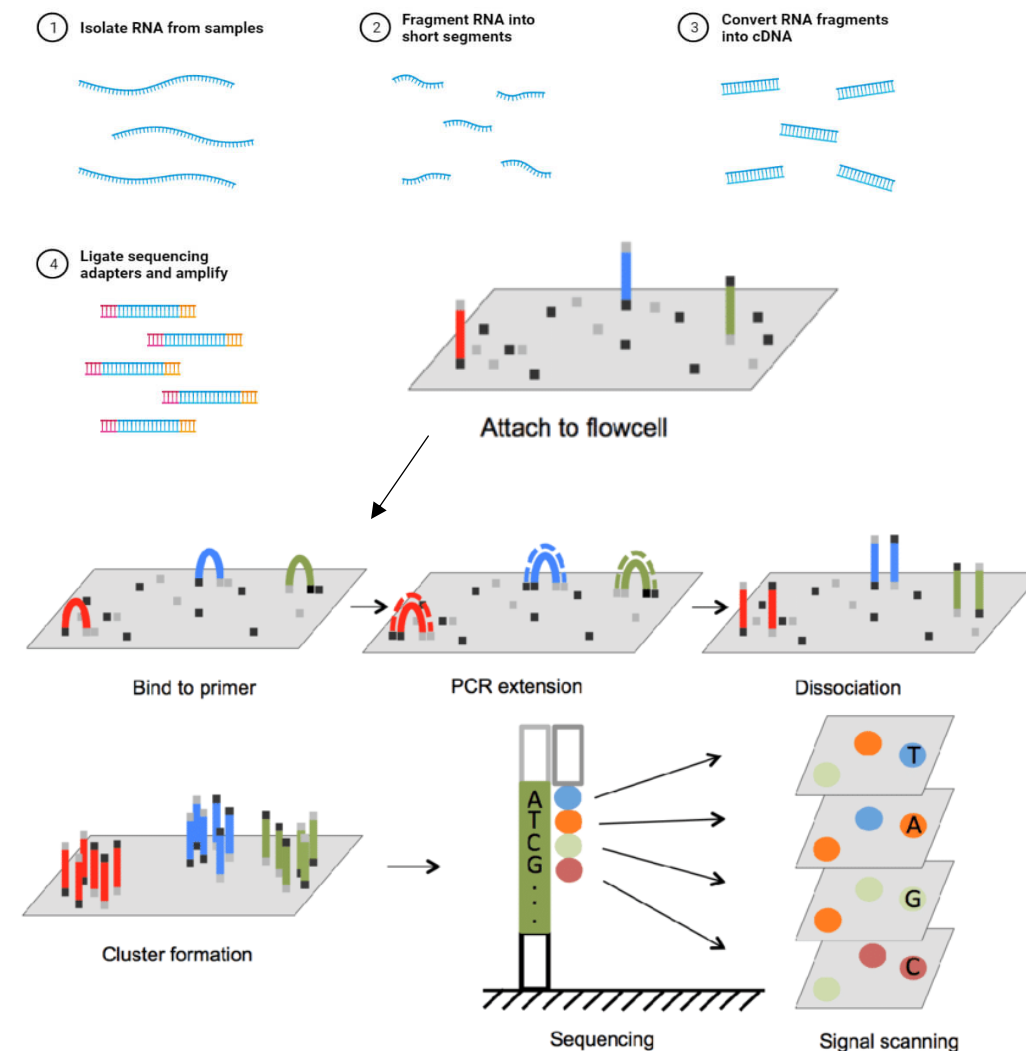


Wet lab - RNA-Seq

RNA sequencing = cDNA sequencing

- 1. Problem: different RNA species
(~ 80-90% of total RNA is ribosomal RNA) but we are often only interested in coding mRNA
- 2. Problem: sequencing platforms are designed for **short DNA** fragments

-> Comes with new opportunities
& challenges!

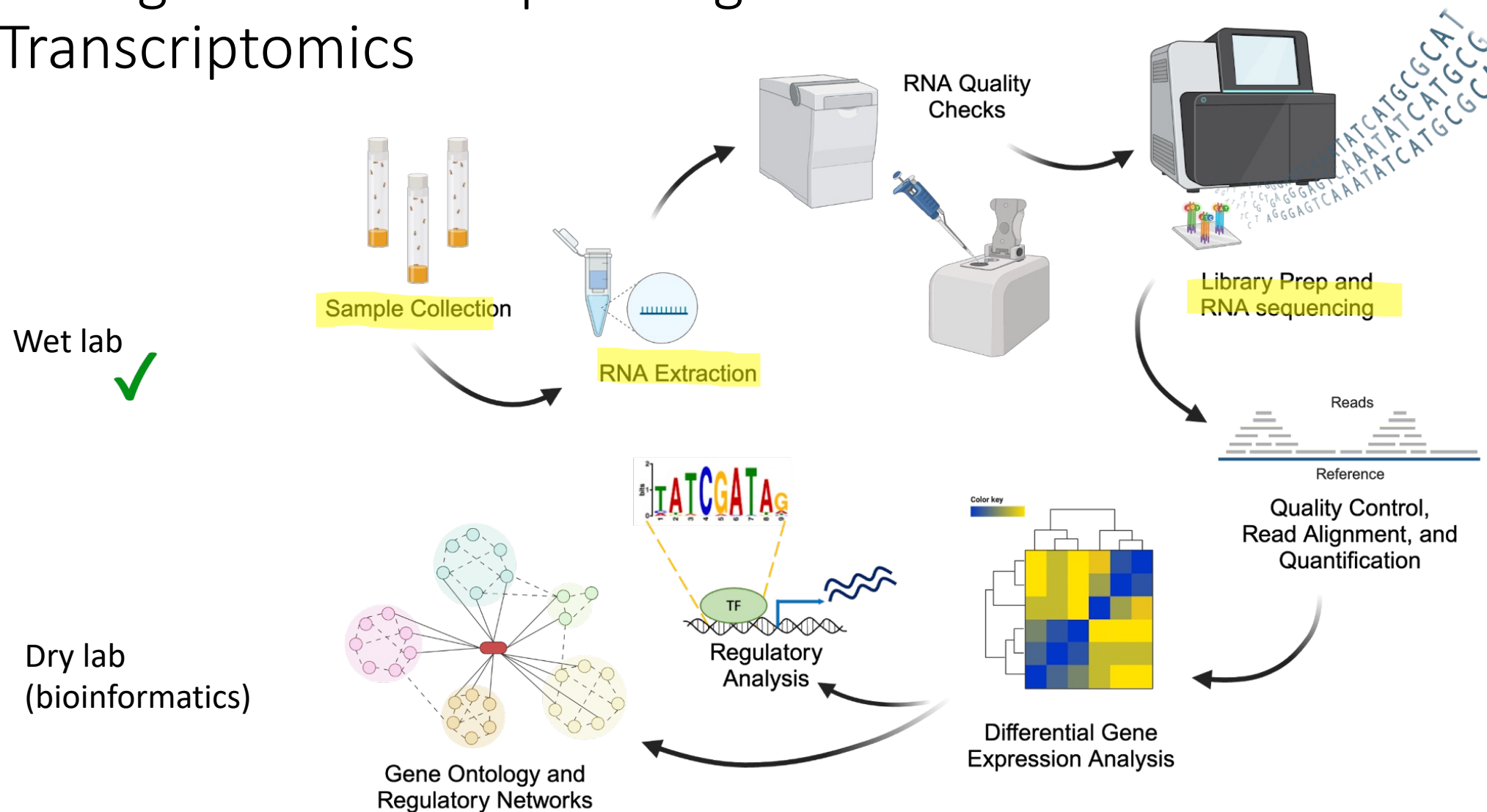


Illumina sequencing

Recap questions

- What are the three steps of a PCR?
- How is the PCR used in different gene expression analysis technologies?
- How do the different gene expression methods differ (& what do they have in common) ?

Next generation sequencing of RNA: Transcriptomics



Installation

- Install these Bioconductor R packages:

```
if (!requireNamespace("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")
```

```
BiocManager::install("DESeq2")
```

```
BiocManager::install("airway")
```

```
BiocManager::install("PCAtools")
```

```
BiocManager::install("apeglm")
```

- Install these standard R packages:

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
install.packages("dplyr")
```

Dry lab - Raw RNA-Seq data

Raw data contains technical artifacts that we
need to account for bioinformatically

Dry lab - Raw RNA-Seq data

@M03234:108:000000000-CFH62:1:1105:22330:22739 1:N:0:1 Header

TCTGGGTGGCCGAAAAATCAAAATAAATGTTGATATAAAACAGGGTCACCGCCTCCAGCTGGGTCAAAAAAAGTAGTATTAAATTCCTATCTGTTAAAAGCATTGTT [Read](#)

+

AFFFFFFAGFGGGGGCGGHFHFHHFHGGGHGFGGDAGHGGEGCHG2AGFHEAFFGHHEFHFHG?FGHHHHGGG@B4@F4GFHHHHHHFHDGEHHGHFHHHGGHDG3BGH [Quality string](#)

@M03234:108:000000000-CFH62:1:1106:13496:12956 1:N:0:1 [Header](#)

TCTGGATGTCCAAAGAACCAAAATAAATGTTGATATAATACAGGGTCGCCTCCTCCTGCAGGGTCAAAAAAGGATGTATTAAAGTTTCTATCTGTAAGTAACATTGTTA [Read](#)

+

CFFFFFFFFGGGGGGGGGGGHHHHHHHHHHHHHHHHHHHHHHHHHHHHHGHGGGGGHHHHHHHHHHHHGHHHHHHHGCGGGHHHHHHHHGHHHHHHHHHHHHHHHHHHHHHHHHHH

Quality string

...

...

...

Dry lab - Raw RNA-Seq data

Raw data contains technical artifacts that we need to account for bioinformatically

$$Q = -10 \log_{10}(P_{\text{error}})$$

$$\rightarrow P_{\text{error}} = 10^{-Q/10}$$

- Quality of base calling

...A C G T ...

...I ? 5 ! ...

$$Q = \text{ASCII code} - 33$$

Quality char	ASCII code	Phred score (Q)	P_{error}	Base call accuracy
I	73	40	1 in 10,000	99.99%
?	63	30	1 in 1,000	99.9%
5	53	20	1 in 100	99%
!	33	0	1 in 1	0%

- Adapter sequences
- Experiment-specific biases

Dry lab - Raw RNA-Seq data

Raw data contains technical artifacts that we need to account for bioinformatically

$$Q = -10 \log_{10}(P_{\text{error}})$$

$$\rightarrow P_{\text{error}} = 10^{-Q/10}$$

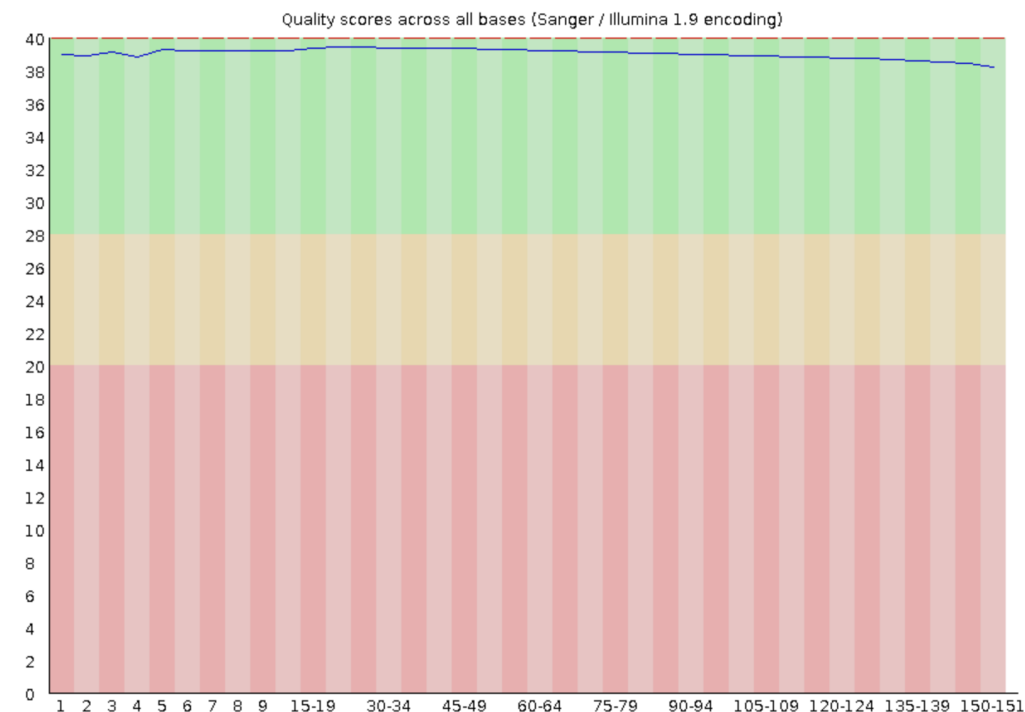
- Quality of base calling

...A C G T ...
...I ? 5 ! ...

$Q = \text{ASCII code} - 33$

- Adapter sequences
- Experiment-specific biases

✔ Per base sequence quality



Dry lab - Raw RNA-Seq data

Raw data contains technical artifacts that we need to account for bioinformatically

$$Q = -10 \log_{10}(P_{\text{error}})$$

$$\rightarrow P_{\text{error}} = 10^{-Q/10}$$

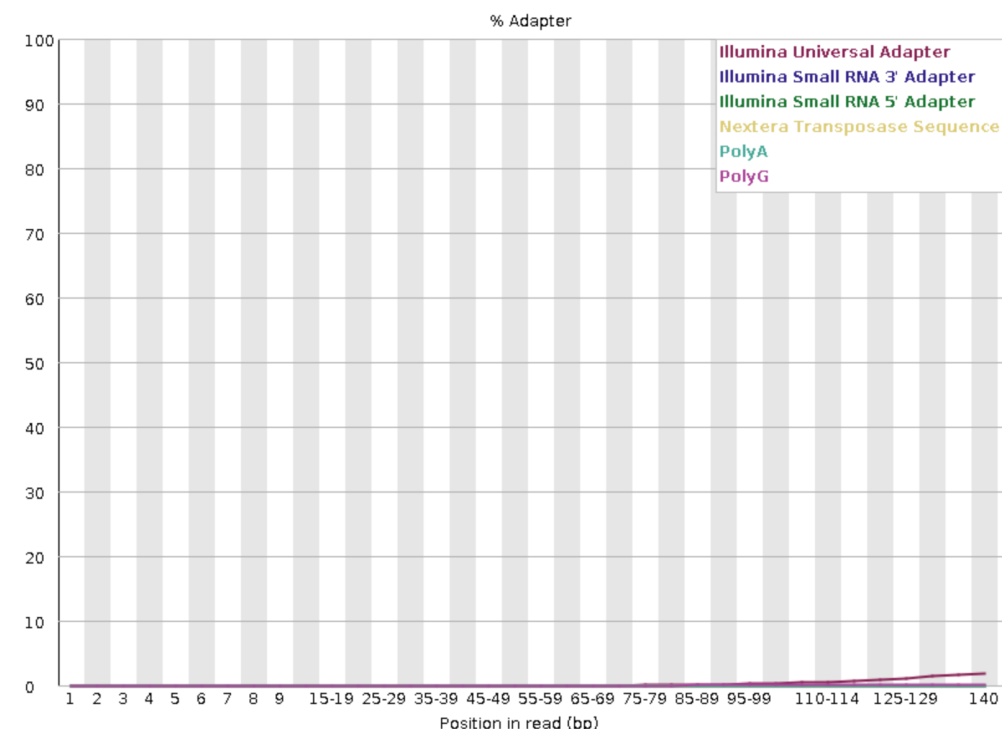
- Quality of base calling

...A C G T ...
...I ? 5 ! ...

$Q = \text{ASCII code} - 33$

- Adapter sequences
- Experiment-specific biases

✓ Adapter Content



Dry lab - Raw RNA-Seq data

Raw data contains technical artifacts that we need to account for bioinformatically

$$Q = -10 \log_{10}(P_{\text{error}})$$

$$\rightarrow P_{\text{error}} = 10^{-Q/10}$$

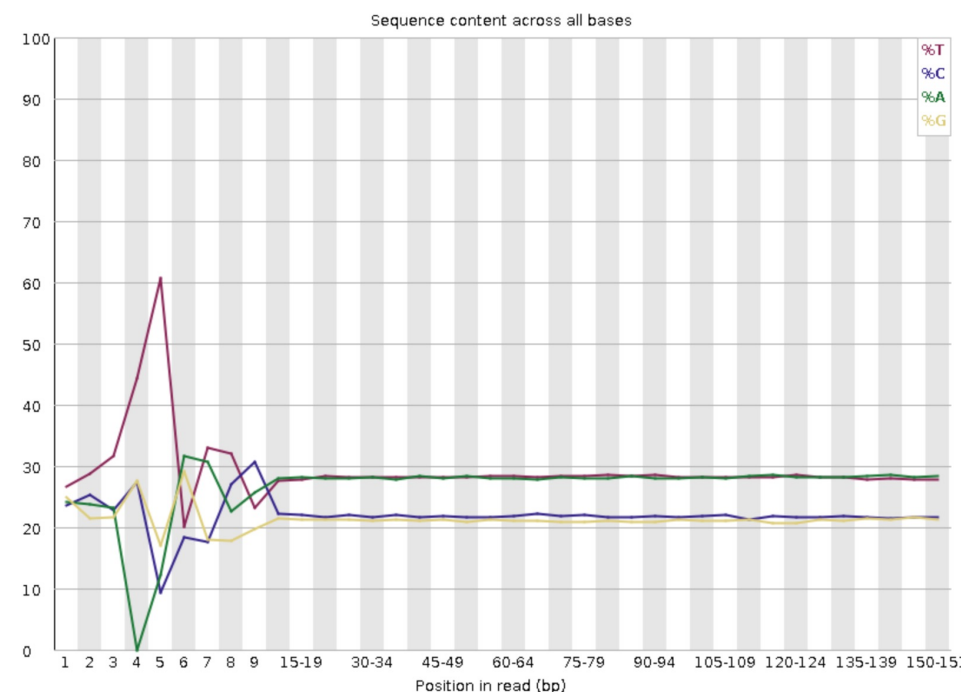
- Quality of base calling

...A C G T ...
...I ? 5 ! ...

$Q = \text{ASCII code} - 33$

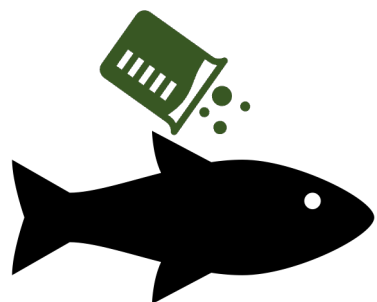
- Adapter sequences
- Experiment-specific biases

✖ Per base sequence content

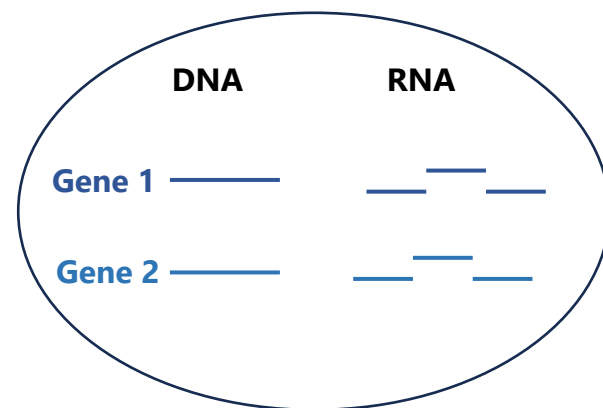
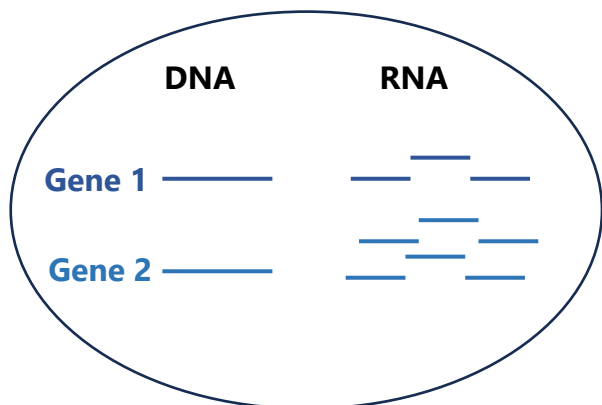
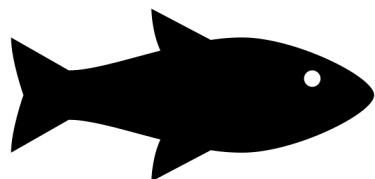


Remember:

The aim is to quantify gene expression changes



Vs.



- cDNA reads correspond to initial mRNA levels

BUT

- Reads (75 – 150 bp) are much shorter than the biological transcript (several kbp)

(Problem not specific for RNA seq, but the resulting challenges are greater than for DNA seq data)

Dry lab - Raw RNA-Seq data

@M03234:108:000000000-CFH62:1:1105:22330:22739 1:N:0:1 Header

TCTGGGGTGGCCGAAAAAATCAAAATAAATGTTGATATAAAACAGGGGTCACCGCCTCCAGCTGGGTCAAAAAAAGTAGTATTAAAAATTCCTATCTGTATAAAAGCATTGTT

+

AFFFFFFAGFGGGGGCGGHFHFHHFHGGGHGFGGDAGHGGECHG2AGFHEAFFGHHEFHFHG?FGHHHHGGG@B4@F4GFHHHHHHFHDGEHHGHFHHHGGHDG3BGH [Quality string](#)

@M03234:108:000000000-CFH62:1:1106:13496:12956 1:N:0:1 Header

TCTGGATGTCCAAAGAACCAAAATAAATGTTGATATAATACAGGGTCGCCTCCTCCTGCAGGGTCAAAAAAGGATGTATTAAAGTTTCTATCTGTAAGTAACATTGTTA [Read more](#)

+

C F F F F F F G G G G G G G G H G H H G G G G G H H H H H H H H H H H G H H H H H H H G G G G H H H H H H H H H H H H H H H H H H H

Quality string

...

...

...

To quantify transcript abundance, millions of short reads are aligned to a reference

- Reference genome -> contains exonic and intronic elements
- Transcriptome -> contains only exonic elements

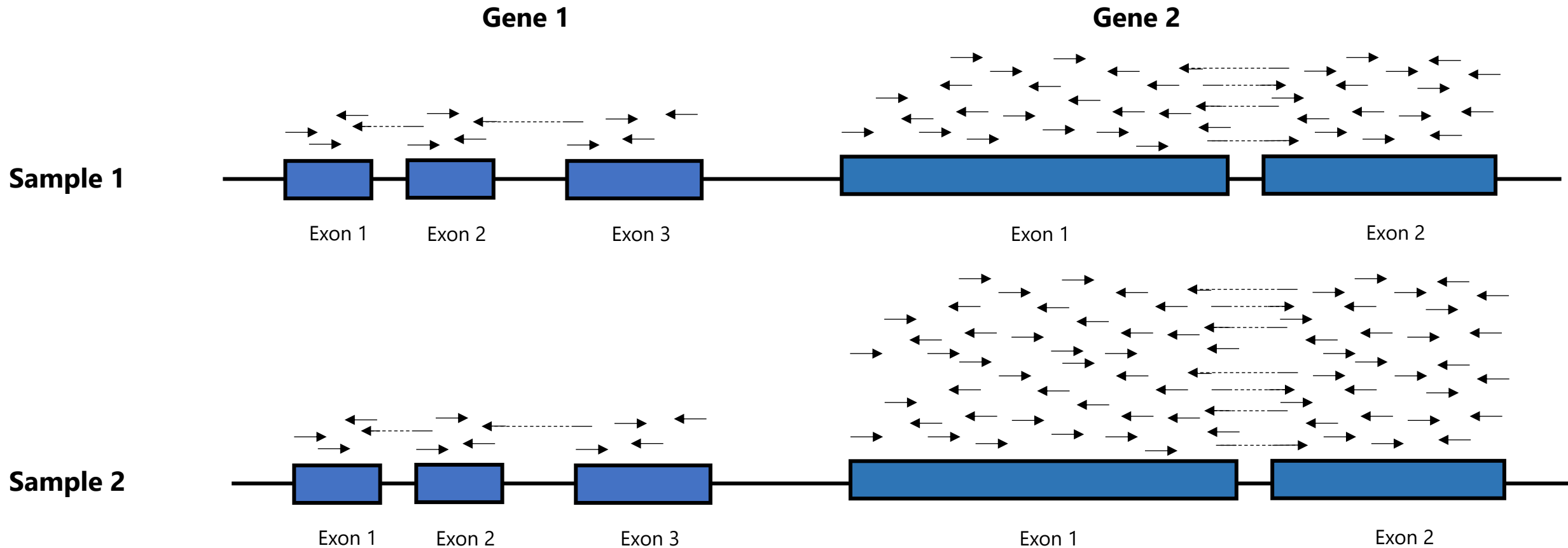
Dry lab - Reference genome

>chromosome1

GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAAGAACATACAAGAAAGATTGTACAACAAAAATCAAGCTTTGATAACGTTACTTAACCGCTGCCATTACCTACCA
CCAACTAGATTTCAGGCCTCAAAACACAATGCAAAACAAGTACAAAAGTTACCTAATATTGCTAGATCACATTCACACGAGACCAAAC
GAAAAGATTTCCTATGCTCTAGAAAAAGCTCTTCCCTACGCCTAAATATTTTCTAATTGTAGGTATGCTAACTTGCTTCTCTATAATTGTAAT
AATAATTAGCAATTGCAATAGGCTAGGGAGATTATCACAGTTACTTCTATTCTAAACACCATCTACGACTTGTATACGAGATAATTAGGTCT
GGAATCAGAAAGAAAAGGTAAATTAAGTAGTCTCCTTGAAATGCAAGCGTGGAAGTACATCAATCTCACAGGAGCTTAGATGAACAC
GAAATAACAGATACCTCGCGGAGAGAAAATAGCCTTTCAAAACGAAAACGCGAGTAAGTACCAAGACACTCTCCTGTCATAGCAAC
GATCTTCAATTTATAAATTCAAATCAATAATGCGGTGCTTGGTCATCAAAGAGAGGACTGGTGAAATTGCCGGACTCAAAGACTCATCCG
TCTTCCGCTCGTATGCACAATTATACACGGTATGATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCA
GGCGTTATCAGGTGTCGCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGT
CAACCGTTCCAGACCAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGACCGGAGCTCATTCAATTAGAT
TAGGTTAAGAGGTCTCGTCAACGGCAGCTTCCGAAGCGAGTCTGCCAGCTCCTCCGTACGGCTCAGTCAATTAGGACTCTTGTAGATGC
ACATACAATTTCTAGCAGAGCTTCTGAGAGCGAGCAGGTAAGTACTAACAAGCCAAGATGACAATGACAGCACAAGCAAGTACTCTTT
TAAGATTAGATTAAGGAGGGTCTGCGCTCCGCCTTAGTGAGCTCCCTCATAGGACTCGACACAACATAGATACTAACACACTTCTATCA
ATGCATCAAATAGGTATCTATAGAAAAGGGTACGCACGCAAATCACTCGGGTCATATACATAACAAAACGACCATTACATCAAAATAGA
CTATCGTACGGTTAACTGCATATTTCTTCGCCAACATTTGTACACGATATGACTTGGATTAACATGGGGGAGCGTTGGGAAAGGTTTTTCAT
CAACTCAGCACTCAAAAATACCTGAATAGTTGTGTGCTATTTTCTACAAGCTATAAATAGCCTAGCGAACTTGTTAGGAGACATCCAACG
ATTAATTTTGAATTGCCTCGTCACAGAACAGCCAATGACTAAACATTGCTCTTTATCTCACC...

Gene 1
(exons/introns)

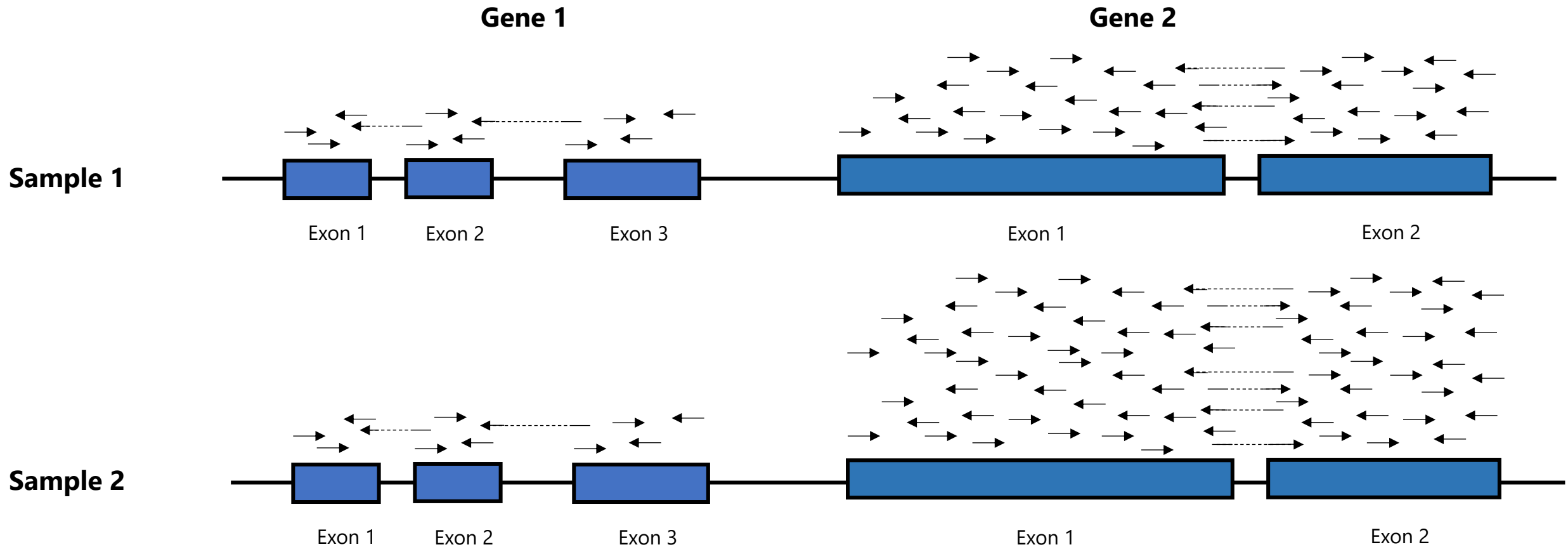
Dry lab - Read alignment to genome



Dry lab - Read alignment

	Sample 1	Sample 2	...
Gene 1	500	650	...
Gene 2	2504	5896	...
...	...	...	...

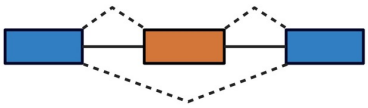
Count matrix used to test
for differential expression
between samples



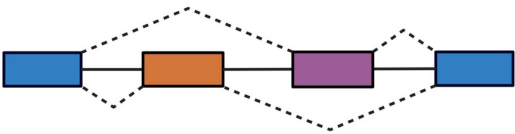
Dry lab - Reference transcriptome

Alternative
splicing

Skipped exon (SE)



Mutually exclusive
exons (MXE)



CS



AS



CS



AS



>Gene1.transcript1

GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAGAAGACATACAAGAAAGATTGTACAACAAAAATCAAGCTTTGATAACGTTACTTAACCGCTGCCATTACCTACCA
CCAAACTAGATTCAGGCCTCAAAACACAATGCAAAACAGGACTGGTGAAATTGCCGGACTCAAAGACTCATCCGTCTTCCGCTCGTATG
CACAATTATACACGGTATGATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCAGGCGTTATCAGGTG
TCGCCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGTCAACCGTTCCAGAC
CAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGAC

>Gene1.transcript2

GTTGCTTCTATTTAAGTTCTTTTTACTTGATATAAATATAGATAAATATATACAAAAGAAATAGTTCAGATAATCGCAAATAATTTTCAGAGC
GAACATTTAAAATGTGTTCTGATCCAAGTTTCACTGTTACTCCTCCTACATATGTTGAACGTTTCCTTGCTTCAGCTCTACGTAGGCATATGC
TCACGAGTAGCACAGAAGACATACAGTATGGAGGCAGGAGAGGACAAAGAACGGGAGGGTAGGAGCTGGCTATCCCATGCAGGCGTT
ATCAGGTGTCGCCAATAGCAATACATGAAAGTCAGTACGGGGTACAAAAACCTTACCCGGAAAATGAAACAGCCTAATACTGTCAACCG
TTCCAGACCAGTCAACTGCAAGATTATTGTCAATTAGCAAATTAGATCAAGAAGTTTATCTCGAC

>Gene2.transcript1

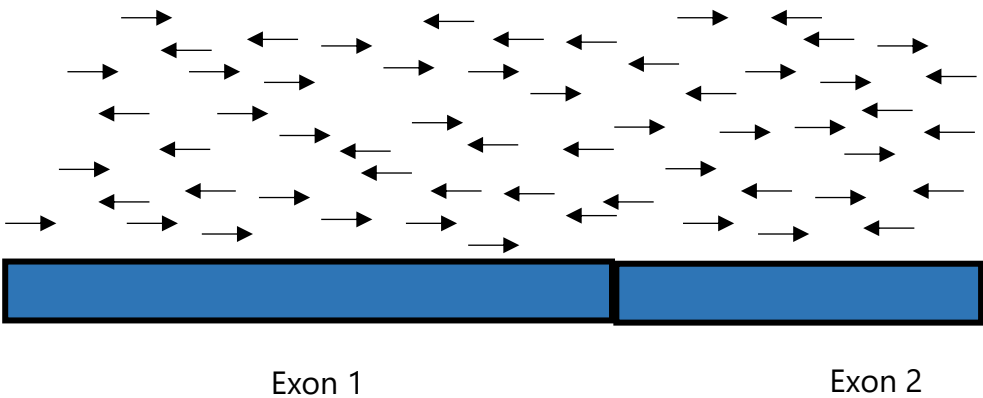
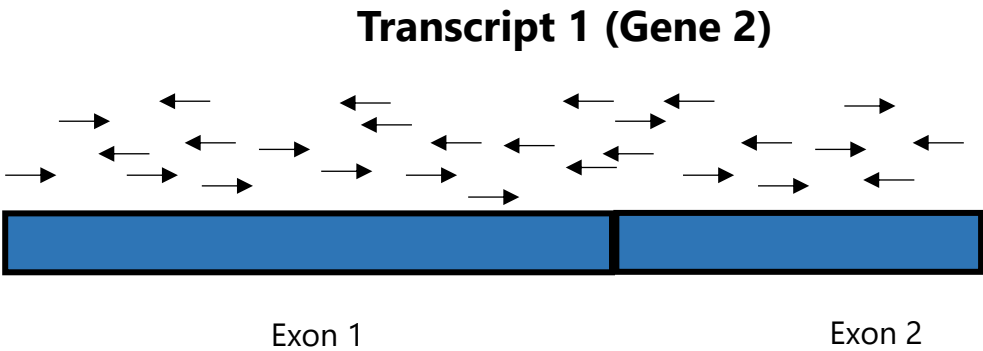
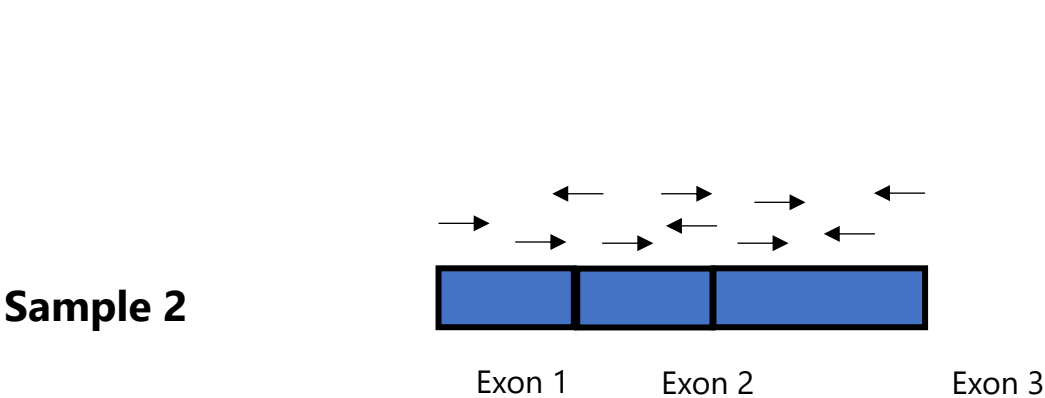
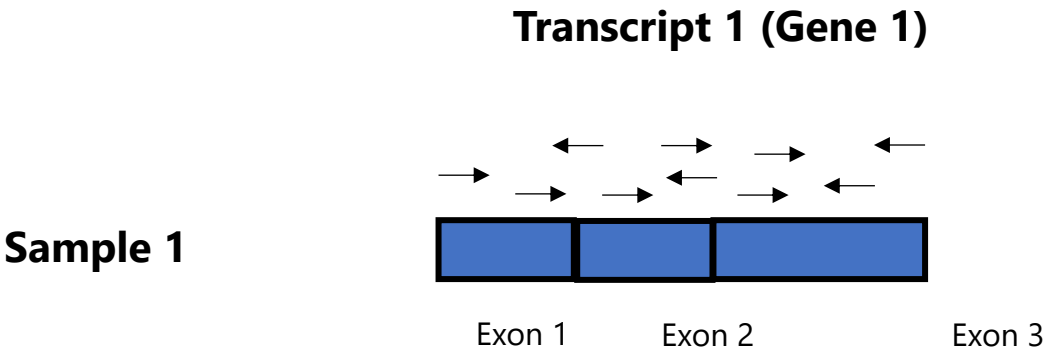
ATTTTCTACAAGCTATAAATAGCCTAGCGAACTTGTTAGGAGACATCCAACGATTAAATTTTGCAATTGCCTCGTCACAGAACAGCCAA
TGACTAAACATTGCTCTTTATCTCACC...

Transcripts of gene 1
(2 alternatively spliced
isoforms)

Transcript of gene 2

Dry lab - Read alignment to transcriptome

	Sample 1	Sample 2	...
Gene1.Transcript1	416	650	...
Gene1.Transcript2	280	300	...
Gene2.Transcript1	2020	4589	...
...	...	...	...

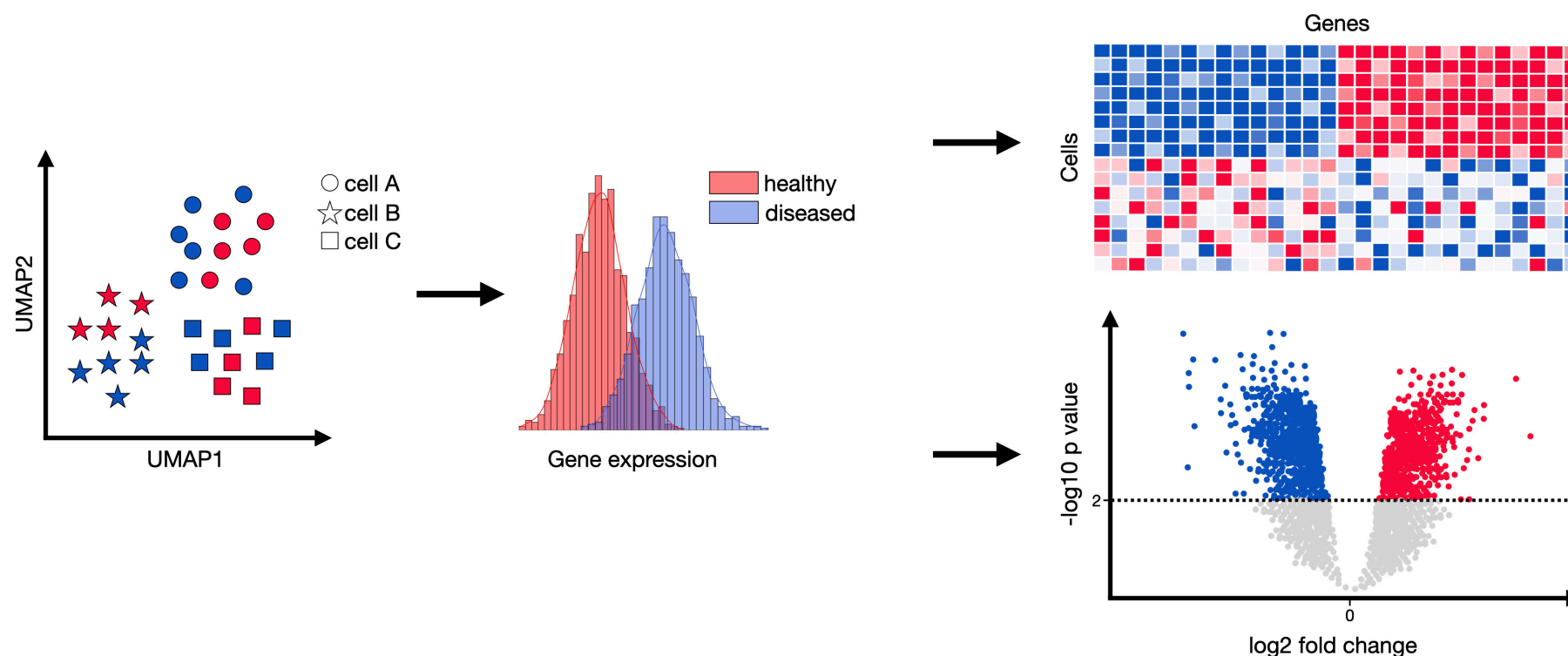


Recap questions

- What are the bioinformatic challenges with short read Illumina data?
- What is a read and how can it be used for expression estimation?
- How do alignment strategies differ between genomic and transcriptomic references?
- What are counts?

Dry lab – differential gene expression analysis

- Test if gene expression patterns (= abundances, counts) differ between sample types



Dry lab – differential gene expression analysis

- Test if gene expression patterns (= abundances, counts) differ between sample types

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
Gene 1	500	650	758	0	10	47
Gene 2	2504	5896	3587	10256	11589	9584
Gene 3	3587	5895	2598	257	894	685
Gene 4	2569	3512	2014	4512	3520	2597
...	...	...	...	...	...	...
Gene 20 145	78	95	63	150	26	34

1. Model counts for each gene:

Counts $\sim$ condition

Modelling count data

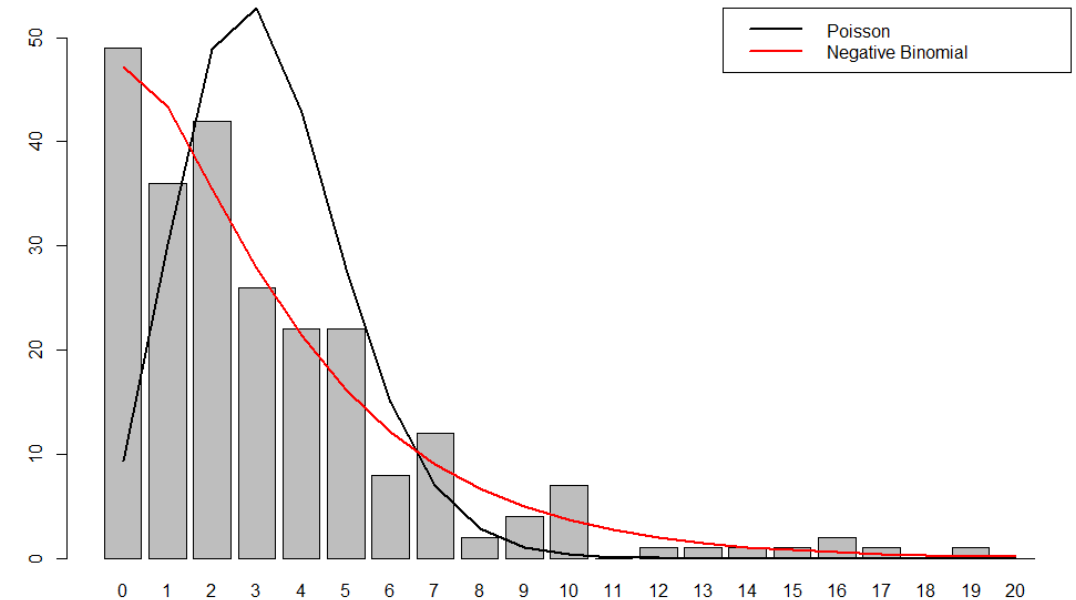
- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene:

Counts $\sim$ condition

- Count data is not normally distributed (Poisson, Quasi-Poisson, Negative Binomial)

Gene counts



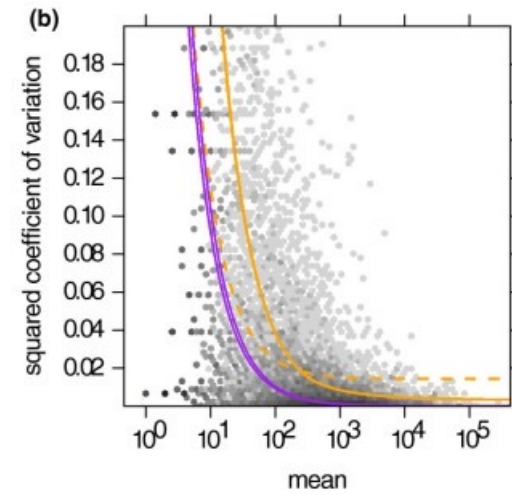
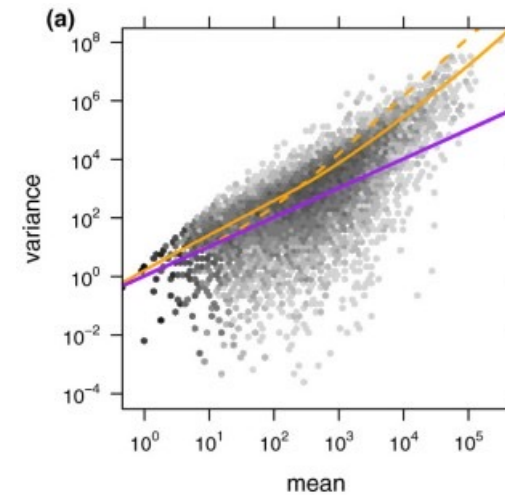
Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene:

Counts $\sim$ condition

- Count data is not normally distributed (Poisson, Quasi-Poisson, Negative Binomial)
- Variance depends on the mean (heteroscedasticity)



Modelling count data

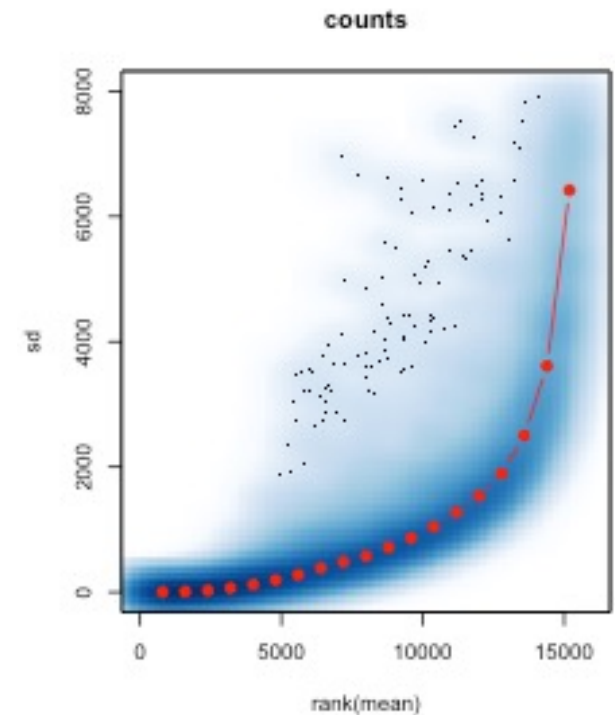
- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene:

Counts $\sim$ condition

- Count data is not normally distributed (Poisson, Quasi-Poisson, Negative Binomial)
- Variance depends on the mean (heteroscedasticity)

Solution: Fit negative binomial GLMs



Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene:

Counts $\sim$ condition

- Count data is not normally distributed
(Poisson, Quasi-Poisson, Negative Binomial)
- Variance depends on the mean (heteroscedasticity)

2. Test for differential expression: hypothesis tested that the LFC between groups $\neq 0$

Modelling count data

- Test if gene expression patterns (= abundances, counts) differ between sample types

1. Model counts for each gene (NB GLMs)
2. Test for differential expression

-> R packages **DESeq2** (Love et al., 2014) & **edgeR** (Chen et al., 2008) most widely used

Recap questions

- What means differential gene expression analysis?
- How is it performed?
- What is important to consider when analyzing/modelling count data?